

Alex Mercer

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PROFESSIONAL SUMMARY

High-impact Software Engineer with 5 years of experience architecting resilient distributed systems and high-throughput backend services. Proven expertise in scaling mission-critical infrastructure, optimizing PostgreSQL performance, and leading cross-functional teams to deliver secure, enterprise-grade solutions. Adept at leveraging Go, cloud-native technologies, and IaC to drive operational efficiency and reliability at scale.

EXPERIENCE

TechNova Solutions

Software Engineer II

Austin, TX

January 2023 – Present

- Architected and migrated a legacy monolithic REST API to a serverless AWS microservices architecture, reducing infrastructure costs by \$150k annually (35% reduction) while supporting 5,000+ requests per second.
- Spearheaded the modernization of the primary customer portal using React and TypeScript, elevating Lighthouse performance scores from 68 to 94 through strategic code-splitting and asset optimization.
- Accelerated deployment velocity by 50% and achieved near-zero production rollbacks by implementing robust CI/CD pipelines with GitHub Actions, Docker, and Terraform for Infrastructure-as-Code.
- Integrated Prometheus and Grafana for real-time observability, reducing mean-time-to-resolution (MTTR) for production incidents by 40%.

DataStream Analytics

Junior Full-Stack Developer

Dallas, TX

June 2021 – December 2022

- Optimized frontend performance for 50,000+ monthly active users by refactoring core components using Vue.js and Vuex state management.
- Optimized PostgreSQL database schemas for high-volume IoT data ingestion (processing 50GB+/day), resulting in a 20% improvement in complex query execution speeds.
- Architected a comprehensive testing suite with Jest and Cypress, achieving 85% branch coverage and ensuring system reliability across the primary repository.

DataStream Analytics

Software Engineering Intern

Dallas, TX

May 2020 – August 2020

- Engineered an internal data visualization dashboard using Python, Flask, and Chart.js to provide real-time visibility into API usage metrics.
- Resolved 40+ critical bug tickets within a legacy PHP codebase, significantly improving system stability and reducing technical debt.

TECHNICAL SKILLS

Languages & Frameworks: Go, JavaScript/TypeScript, Python, React, Node.js

Infrastructure & Tools: AWS (Lambda, EC2, S3), Docker, Kubernetes, Terraform, PostgreSQL, Prometheus, Grafana, CI/CD

Core Competencies: Distributed Systems, System Architecture, Performance Optimization, IaC

PROJECTS

CacheFlow - Distributed Key-Value Store | Go, gRPC, Docker, Redis

Engineered a high-performance, distributed in-memory key-value store in Go, implementing the Raft consensus algorithm to ensure fault tolerance and data consistency. Benchmarked to sustain 15,000+ concurrent reads/sec.

DevSync Collaboration Tool | React, Node.js, Socket.io, MongoDB

Developed a real-time collaborative code editor featuring multi-cursor synchronization and live chat. Optimized WebSocket communication to maintain sub-50ms latency.

PUBLICATIONS

Scaling Node.js Microservices Under Heavy Load | Medium - Better Programming

Authored a technical deep-dive on scaling strategies for Node.js, covering clustering, reverse proxy configurations, and multi-layer caching architectures.

EDUCATION

University of Texas at Austin

Bachelor of Science in Computer Science

GPA: 3.8. Focused on Distributed Systems and Cloud Computing.

Austin, TX

August 2017 – May 2021

CERTIFICATIONS & AWARDS

AWS Certified Developer - Associate (2023); **Certified Kubernetes Administrator (CKA)** (2024)

Innovation Award - Q3 2023 (TechNova Solutions); **1st Place - FinTech Hackathon** (Texas Tech Hub, 2022)